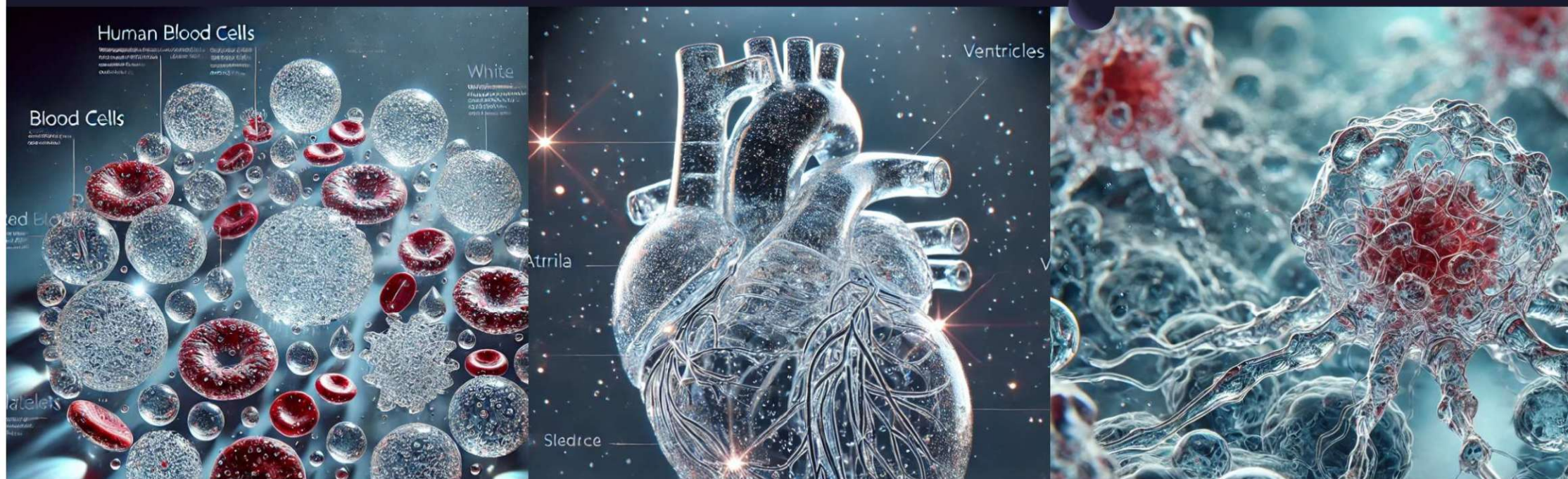


RSs in Metagenomic population studies

Ghith et al (2022).





Sustainability Development Goals and biomarkers of Antimicrobial Resistance: differential abundance analyses

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Revised abstract

Background:

Inequality in global health outcomes such as antimicrobial resistance (AMR) is shaped by macro determinants of health within and between countries.

Therefore, this study aims to assess the potential relationship between the country's development level - progress towards achieving the sustainability development goals (SDG) and the burden of AMR.

The study identifies resistance genes, phenotypes (antibiotics), classes of antimicrobial resistance and bacterial taxa that are differentially abundant between the sampling sites within countries, and compliance with SDGs goals that might explain variation in the global burden of AMR between countries.



Methods:

The Global Sewage Surveillance project consortium collects untreated sewage samples to assess the global burden of AMR. Samples were sequenced to obtain data on the microbial compositions within site-country covering information on resistance genes and bacterial taxa. Samples were collected between 2016 and 2018 and came from 233 cities (102 countries).

A combination of models is developed to describe and characterize the data on AMR in association with SDG indices and goals. A number of SDG indices were added to the analyses to evaluate the study aim. Data on SDGs indices were extracted from the World Bank and the Global Burden of Disease data portals.

Results:

The microbial communities across sites differ from one another, which are attributed to site-country development levels (e.g., measured with SDG index and goals). The study identified a number of differentially abundant taxa at different levels of the bacterial taxonomy as well as differentially abundant genes, phenotypes and classes of AMR.

Conclusions:

Further interventions focusing on improving adherence to SDG goals have the potential for elevating inequality in the global burden of AMR. Yet, a complementary assessment for additional sampling locations is required to get data that are more granular on the global and local burden of AMR and to determine the most appropriate course of action.

Methods

Metagenomics samples were sequenced to obtain information on 19 classes of antimicrobial resistance, 65 phenotypes, 562 resistance genes and bacterial taxa.

To assess the study aim, the findings were triangulated using the wilcoxon-mann-whitney test (WMW), negative binomial models (NB), random effect models, compositional data analysis, and analysis of compositions of microbiomes with bias correction (ANCOM-BC or ABC).

Applying random effect models, the contextual effect of the city was quantified using the variance partition coefficient (VPC) by adding the random effects of the city in the analyses.

A number of SDG indices were added into the models to evaluate the study aim. Data on SDGs indices were extracted from the World Bank and the Global Burden of Disease data portals. The model variance was estimated as the median of the posterior distribution obtained by the Markov Chain Monte Carlo (MCMC) method.

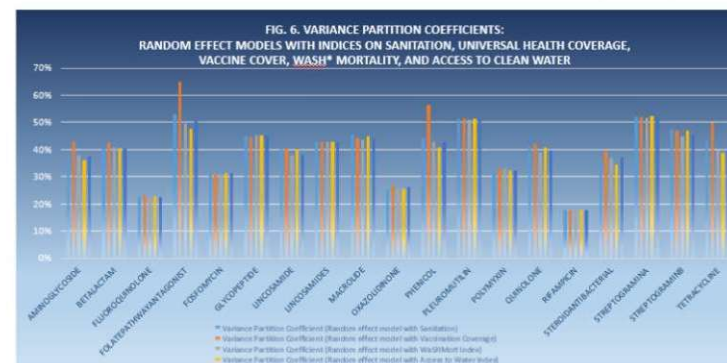
The work is supported by The Novo Nordisk Foundation (NNF16OC0021856: Global Surveillance of Antimicrobial Resistance). Photos credit: unsplash website



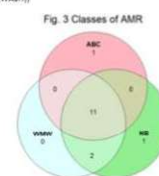
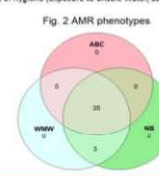
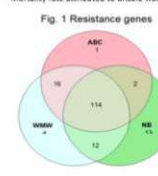
*The index covers all the 17 goals



*The index covers 41 health indicators



*Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe Water, Sanitation and hygiene for All (WASH))



Results

The microbial communities across sites differ from one another, which is attributed to site-country development levels (e.g., measured with SDG index and goals). The study identified 114 resistance genes, 35 phenotypes and 11 classes (Figs. 1-3) that are differentially abundant between countries.

Classes of AMR that have the smallest p-values across the random effect model analyses are: Aminoglycoside, Folate pathway antagonist, Lincosamide, Macrolide, Oxazolidinone, Phenicol, Pleuromutilin, Polymyxin, Streptogramin-A, Streptogramin-B, and Tetracycline.

Adding the SDG global index and SDG health scores into the random intercept models explained around 18%-52% of the between city variation in total AMR (Figs 4 and 5). Similar findings obtained by adding the indices on Sanitation, Universal Health Coverage (UHC), Vaccine Cover, WASH Mortality, and Access to water individually into the models. The variance partition coefficients (VPC) values are between 18%-65% (Fig 6).

In random effect models (Figs 4-6), considering the SDG indices, the largest reduction in between city variation in the burden of AMR occurred for folate pathway antagonist, pleuromutilin and streptogramin-A classes, and the least reduction (still quantitatively large) was for fluoroquinolone, oxazolidinone and rifampicin classes.

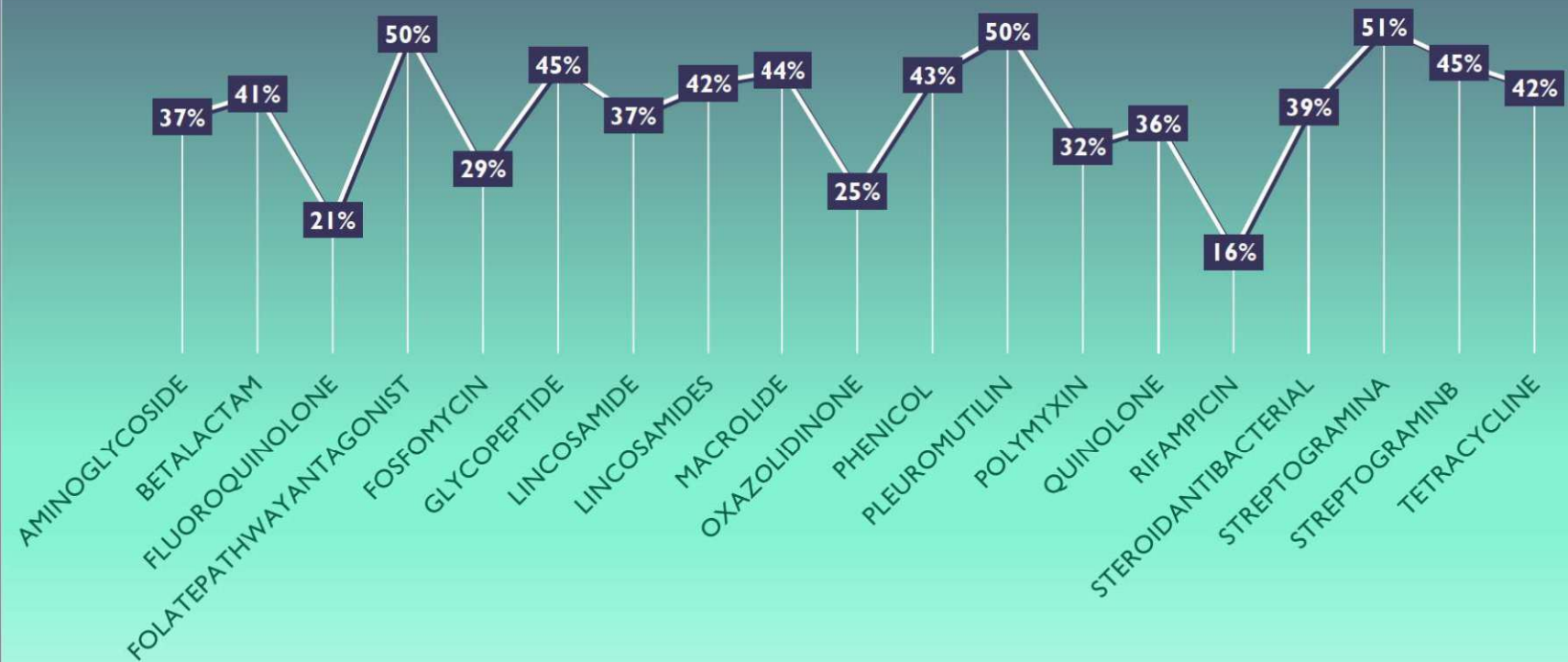
Conclusions

Actions focusing on sustained compliance with the SDG goals have the potential for elevating inequality in the global burden of AMR. There is a need for additional data that cover more sampling locations to develop more comprehensive global monitoring of AMR, and to determine the most appropriate course of action.

Own data

Confidential

FIG. 1. EXPLAINED VARIANCE:
RANDOM EFFECT MODEL WITH
SDG GLOBAL INDEX SCORE*



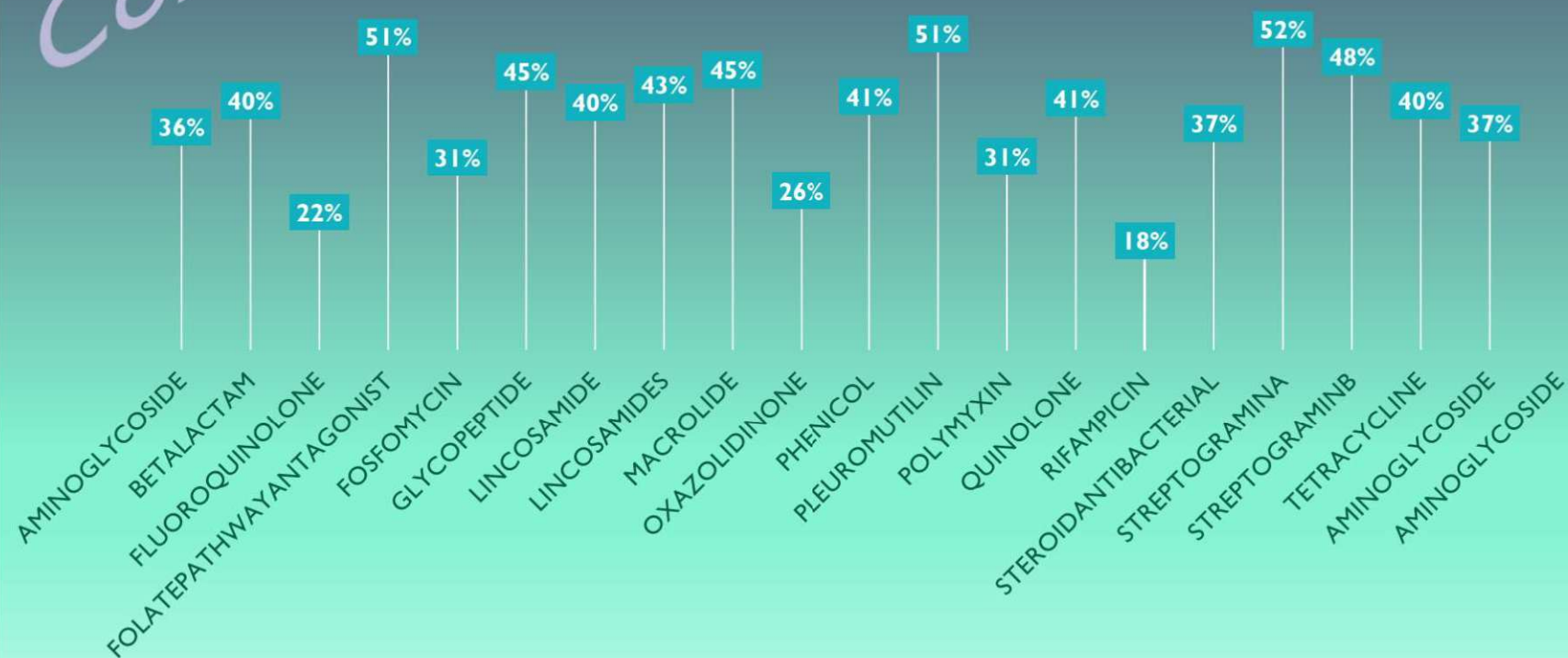
*The index covers all the SDGs 17 goals



Own data

Confidential

FIG. 2. EXPLAINED VARIANCE:
RANDOM EFFECT MODEL WITH
SDG **HEALTH INDEX***



*The index covers 41 health indicators (health index)



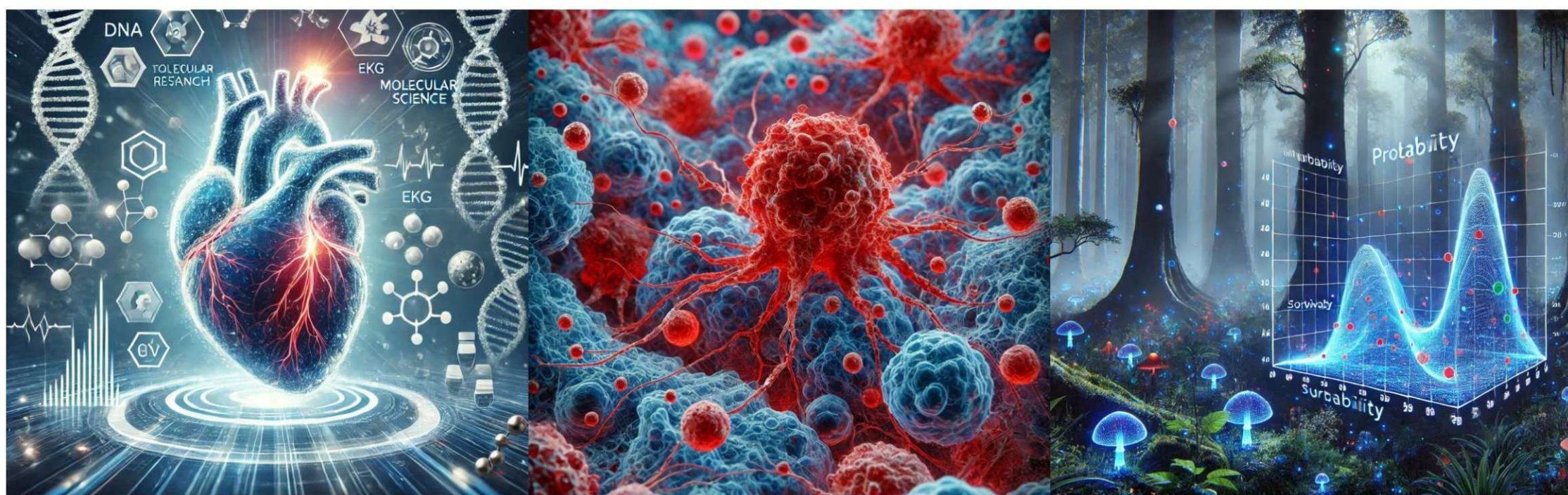
Own data

FIG. 3. EXPLAINED VARIANCE:
RANDOM EFFECT MODELS WITH INDICES ON SANITATION, UNIVERSAL HEALTH
COVERAGE, VACCINE COVER, WASH* MORTALITY, AND ACCESS TO CLEAN WATER



* Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe Water, Sanitation and Hygiene for All (WASH))





Polygenic RSs in prevention

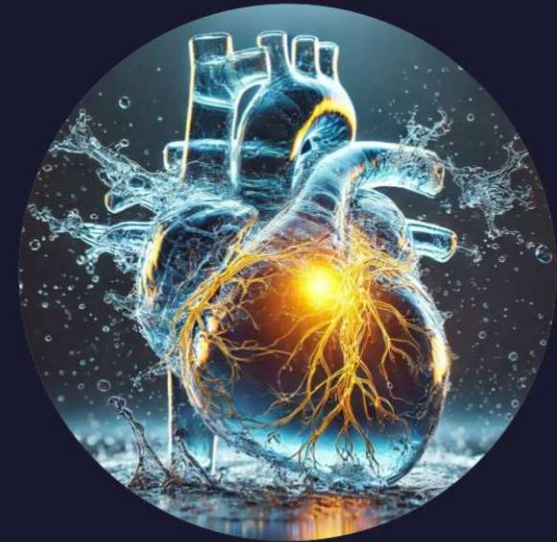
- **Incorporating non-genetic risk factors into polygenic risk score models** - Source: <https://doi.org/10.1038/s41598-023-27637-w>
- **Genotype Score in Addition to Common Risk Factors for Prediction of Type 2 Diabetes** – Source : DOI [10.1056/NEJMoa0804742](https://doi.org/10.1056/NEJMoa0804742)

Polygenic Risk Score

Many people have an illness, or several illnesses, that are affected by **changes** in either one or many of their **genes**, frequently coupled with **environmental** factors.

Researchers are studying these **changes** to understand the role that genetics plays in **diseases** across different populations.

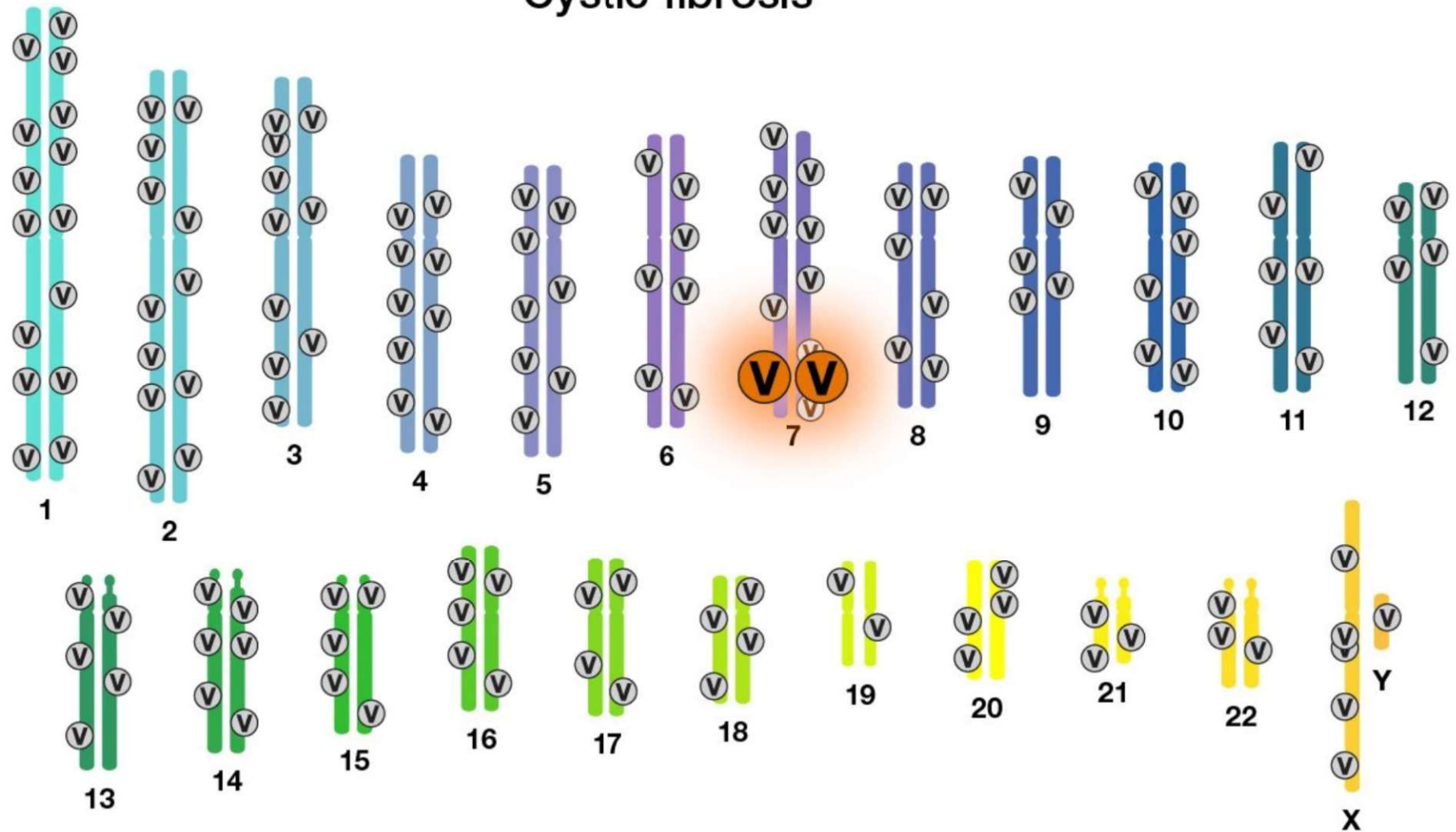
A “**polygenic risk score**” is one way by which people can learn about their risk of developing a disease, based on the total number of **changes** related to the disease.



Single-gene diseases

Cystic fibrosis

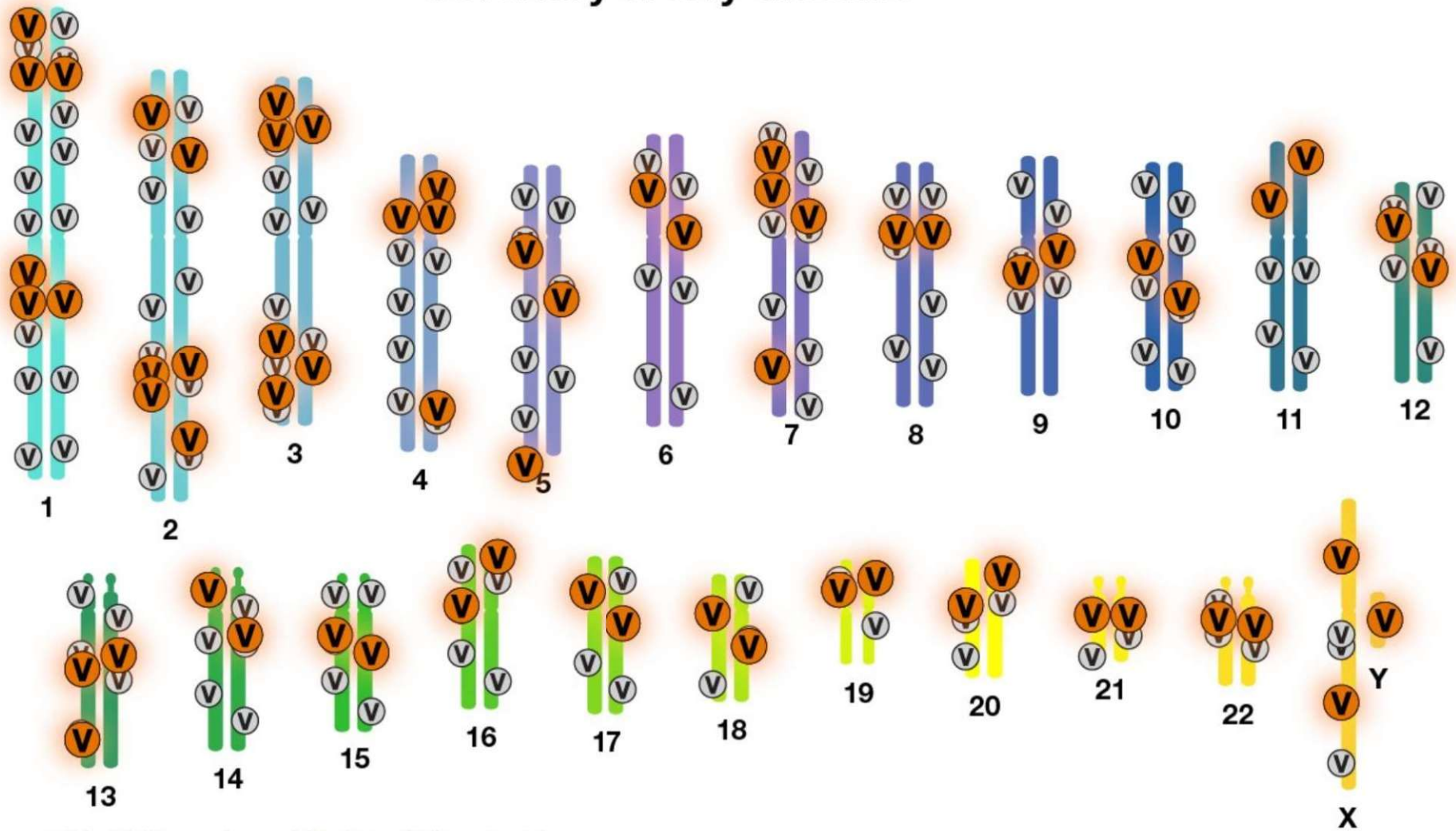
Person
one



Complex diseases - polygenic

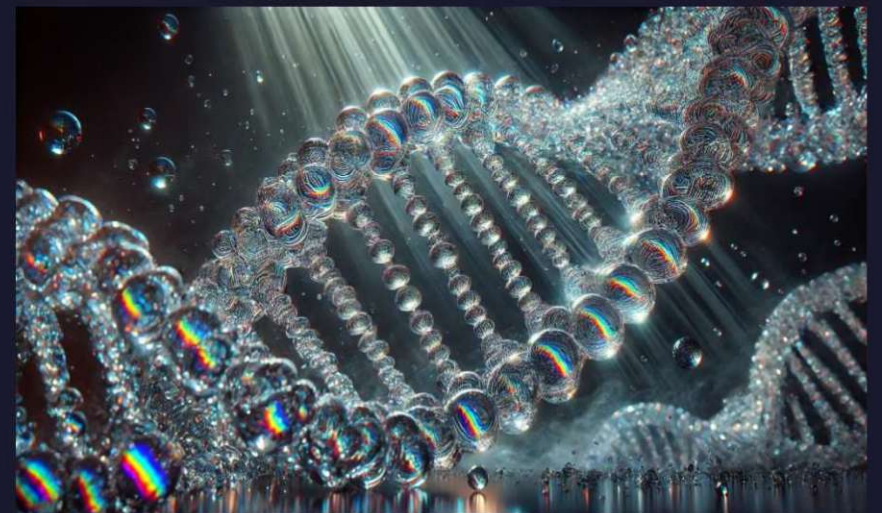
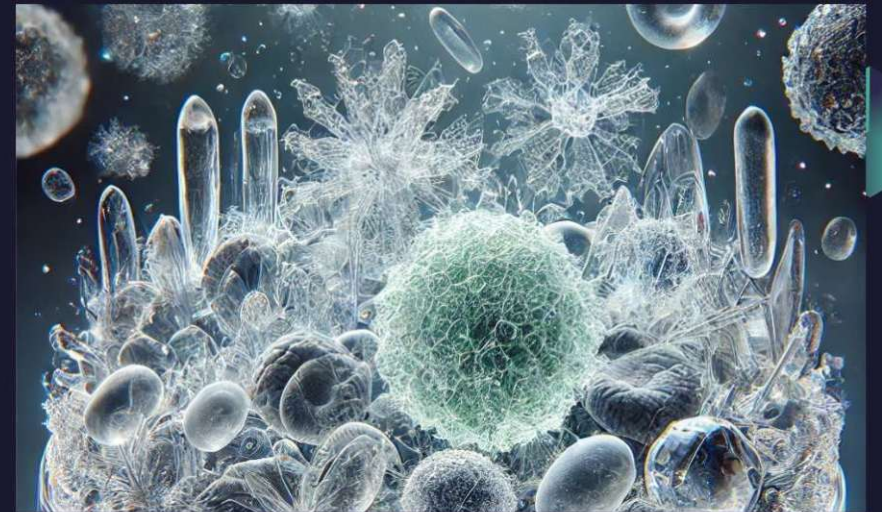
Coronary artery disease

Person
two



The necessity of incorporating non-genetic risk factors into polygenic risk score models

- **Commercial genetic risk scores** - misleading as they overlook easily accessible risk factors like **sex**, **BMI**, **age**, **smoking habits**, **family history**, and **physical activity**.
- **models using UK Biobank data** and externally testing them in the **Lifelines cohort**.
- Incorporating **common risk factors** improves the ability to identify the **top 10% of high-risk** individuals for **type 2 diabetes (T2D)** and **coronary artery disease (CAD)**.
 - Age, BMI, physical activity, sex, parental disease and smoking status
- Comparing different models (**genetics-based**, **risk factor-based**, and **combined**),
 - the incidence of **T2D** in the highest risk group increases from 3.0-fold and 4.0-fold to 5.8-fold.
 - **CAD**, risk increases from 2.4-fold and 3.0-fold to 4.7-fold when using the combined model.
- **Genetic Risk assessment should incorporate additional variables** instead of relying solely on commercial genetic tests.



Genotype Score in Addition to Common Risk Factors for Prediction of Type 2 Diabetes

Background:

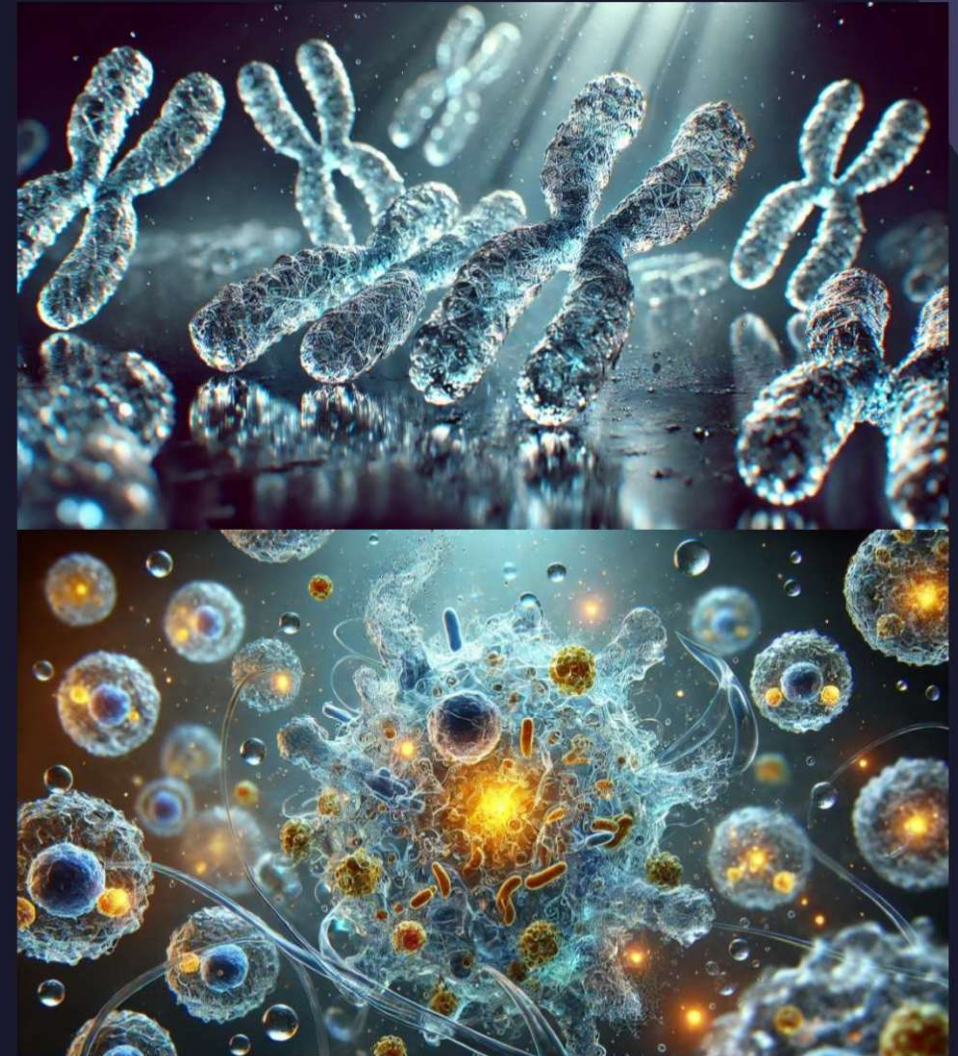
- The hypothesis suggests that understanding **genetic loci linked to type 2 diabetes risk** offers **better risk prediction** than relying solely on **common phenotypic risk factors**.

•age, sex, family history, body-mass index, fasting glucose level, systolic blood pressure, high-density lipoprotein cholesterol level, and triglyceride level

- The **genotype** score led to **risk reclassification** for, at most, **4% of participants**.

Conclusions:

- A **genotype score incorporating 18 risk alleles** successfully identified new diabetes cases in the community.
- However, it **only slightly** improved risk prediction compared to using **common risk factors alone**.



From algorithm to bedside

Global Burden of Disease, Health Analytics, and Access to Therapeutics

- Pandemics influence RCTs
- Market size influences the design and scope of RCTs, as well as access to therapeutics





From algorithm to bedside

Economic evaluations

- Traditional RCS vs omics-based RCTs ??



Evidence pyramid and analytics

Machine Learning vs. Statistics



Feature	Statistics	Machine Learning
Goal	Explain relationships, hypothesis testing, and inference	Predict outcomes and optimize models for accuracy
Focus	Understanding causality , estimating parameters, deriving meaning from data	Pattern recognition, generalization , and automation
Causality vs. Correlation	causal inference (e.g., experimental design)	focuses on correlations – difficult to identify ‘why’
Data Needs	Works well with small , structured datasets	Handles massive, unstructured datasets (images, text, etc.)
Data Abundance	Structured manageable size data	Big data : video, audio, text, images,
Methods	Regression models, hypothesis testing, confidence intervals	Neural networks, decision trees, support vector machines
Complexity	Relatively simple models, interpretable results	Deep learning models with millions of parameters
Algorithmic Advances and complex data structures	struggle with high-dimensional data	ML techniques (e.g., random forests, deep learning)
Feature Engineering	Manual	Automated with e.g., Deep learning methods, convolutional networks in image recognition
Black Box Models vs. Interpretability	Interpretable	Sometimes like " black boxes "
Non-parametric vs. Parametric Models	Usually assume a distribution	often uses non-parametric
Computational power	Classical computation	GPUs , TPUs, and distributed computing

Evidence pyramid and analytics

Common statistical tests in RCTs

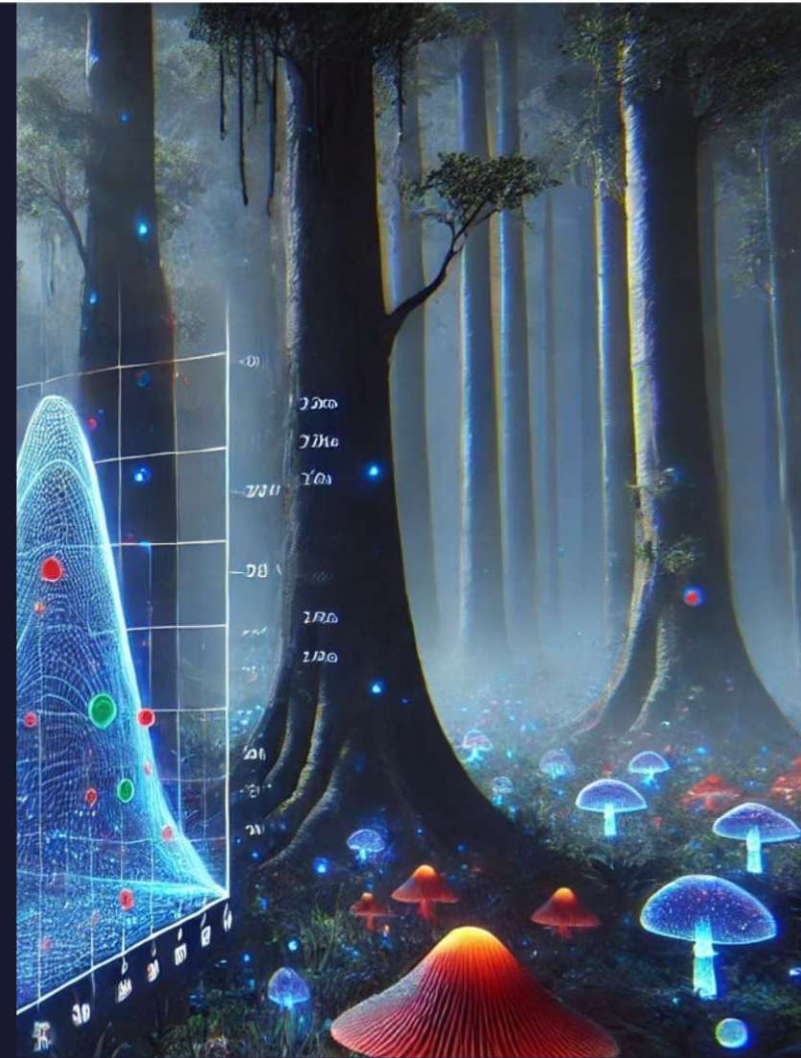


Figure 1: Statistical methods for comparing two independent groups or samples

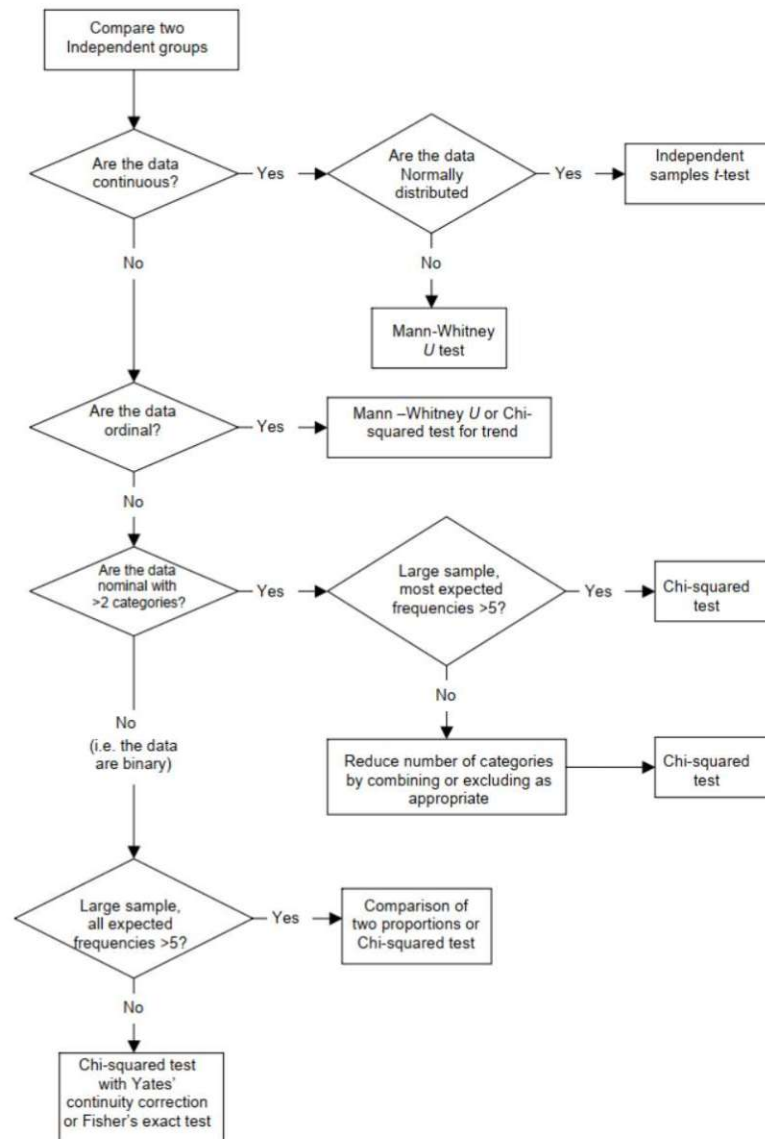


Figure 2: Statistical methods for differences or paired samples

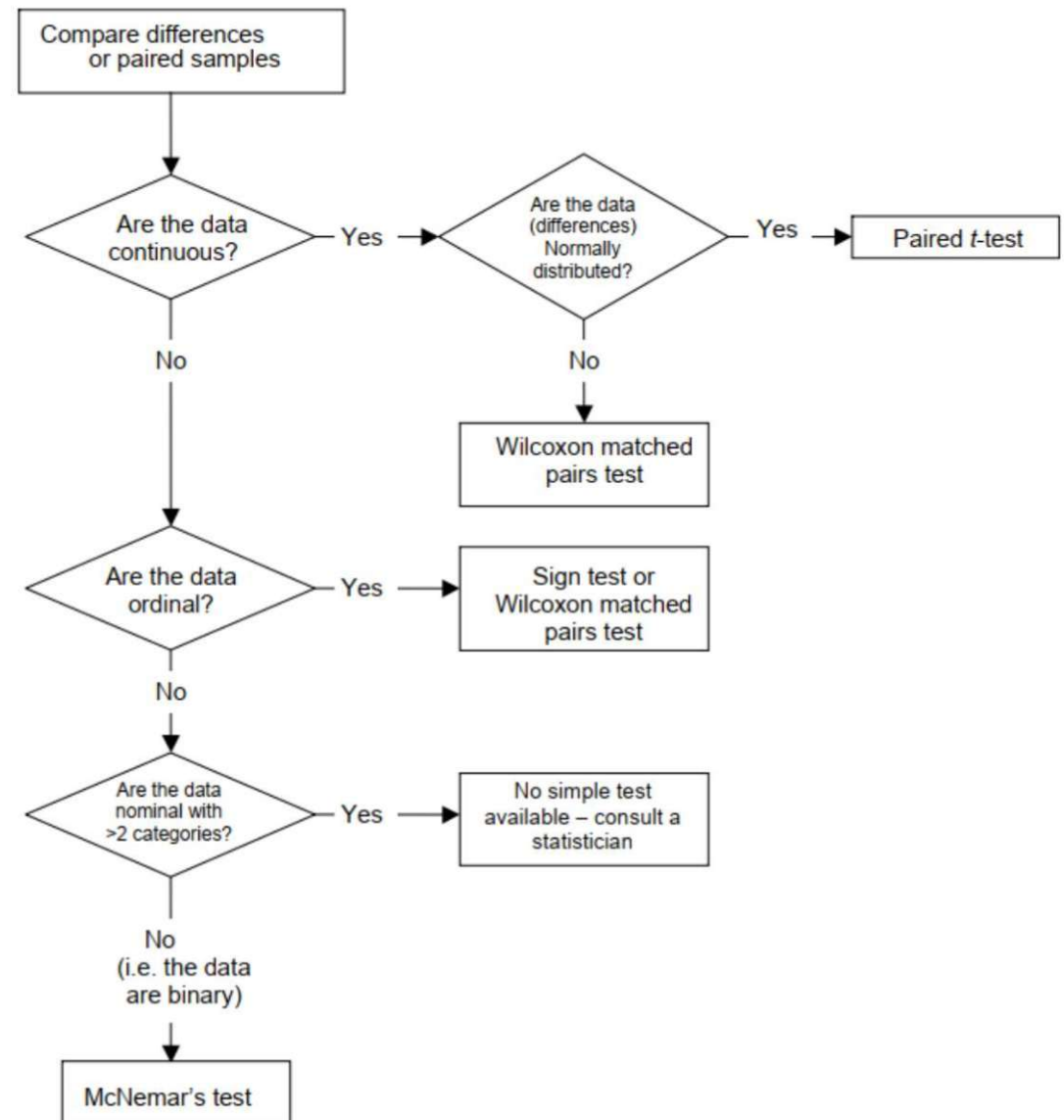
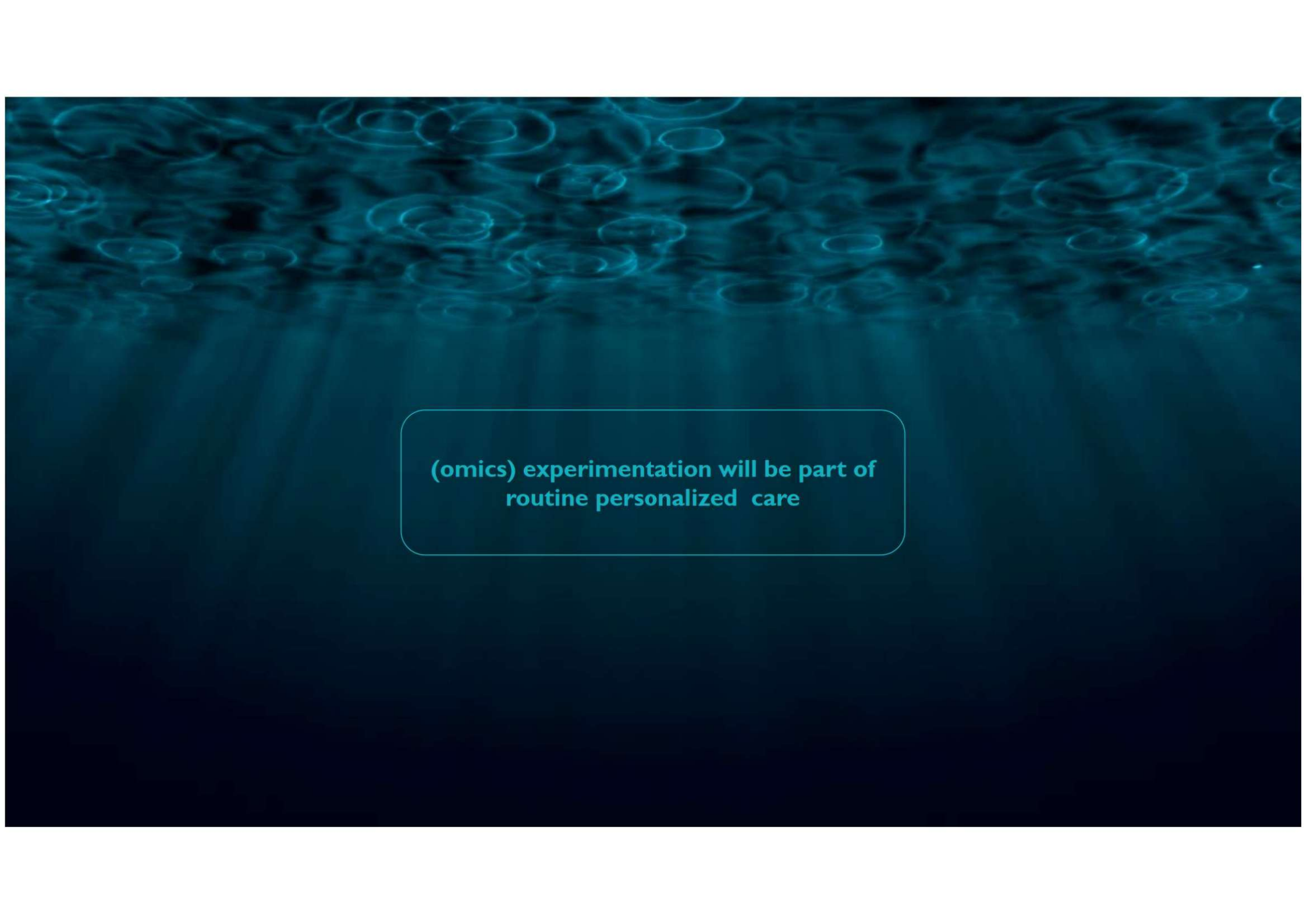


Table 1: Statistical methods for relationships between two variables measured on the same sample of subjects

	<i>Continuous, Normal</i>	<i>Continuous, non-Normal</i>	<i>Ordinal</i>	<i>Nominal</i>	<i>Binary</i>
<i>Continuous</i>	Regression Correlation: (Pearson's r)	Regression Rank Correlation: (Spearman's r_s)	Rank Correlation: (Spearman's r_s)	One-way Analysis of Variance	Independent samples t -test
<i>Continuous, non-Normal</i>		Regression Rank Correlation: (Spearman's r_s)	Rank Correlation: (Spearman's r_s)	Kruskall-Wallis test	Mann-Whitney U test
<i>Ordinal</i>			Rank Correlation (Spearman's r_s)	Kruskall-Wallis test	Mann-Whitney U test Chi-squared test for trend
<i>Nominal</i>				Chi-squared test	Chi-squared test
<i>Binary</i>					Chi-squared test Fisher's exact test



**(omics) experimentation will be part of
routine personalized care**



Thank You

Linkedin connect!

